A VISUAL STUDY OF E-COMMERCE ORDERS, DELIVERY & CUSTOMER SATISFACTION

DATA VISUALIZATION DESIGN (BSCS4001)

PROJECT TECHNICAL REPORT

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1. Executive Summary

This project analyzes the Brazilian Olist e-commerce dataset to identify the operational factors most strongly associated with customer dissatisfaction and to translate those findings into practical actions for marketplace management.

The analysis covers data preparation, exploratory analysis, delivery performance, regional logistics, seller performance, product and pricing characteristics, and recommended interventions. The central finding is that delivery-promise reliability is a stronger customer-experience signal than delivery speed alone. Among delivered and reviewed orders, late orders represented approximately 6.7% of deliveries but generated approximately 32.5% of negative reviews. The negative-review rate was approximately 9.2% for orders delivered on time and 62.3% for orders delivered late.

The results support four priorities:

1. Improve the accuracy and reliability of promised delivery dates.
2. Detect and intervene on late orders before the customer experience deteriorates.
3. Prioritize sellers and routes that generate avoidable dissatisfaction after accounting for order volume and operating context.
4. Use product, freight, and listing-quality information to complement, rather than replace, delivery interventions.

The findings are associative. They identify where operational attention should be concentrated, but they do not by themselves prove that a particular intervention will cause a reduction in negative reviews.

2. Business Context and Objectives

The marketplace must balance growth, seller participation, logistics performance, and customer trust. A high order volume is not sufficient if late deliveries and poor seller experiences create avoidable complaints, refunds, or customer attrition.

The project addresses the following questions:

- How large is the customer-experience problem across the marketplace?
- Does missing the promised delivery date hurt more than taking longer overall to deliver?
- Which regions and seller-to-customer routes show elevated delivery or review risk?
- Which sellers create disproportionate dissatisfaction relative to their order volume and context?
- Which product, freight, and listing characteristics are useful secondary intervention levers?
- How should the marketplace prioritize operational action?

3. Data Sources

The project uses the Brazilian E-Commerce Dataset, supplemented by the repository's derived files and analysis outputs. The raw tables include orders, reviews, customers, sellers, products, order items, payments, geolocation, marketing-qualified leads, and closed deals.

The principal analytical source is `processed_data/master_orders.csv`, which is designed to contain one row per order and combines order status, dates, delivery measures, review measures, item aggregates, payment aggregates, customer information, seller information, and product/category information.

4. Data Preparation

Data preparation is implemented in `preprocess.py`. The pipeline reads raw CSV files from `Dataset/E-Commerce Dataset` and writes reusable tables to `processed_data`.

4.1 Cleaning and standardization

- Date columns are parsed with invalid values converted to missing values.
- Order status values are stripped and normalized to lowercase.
- City names are stripped and converted to title case.
- Product category names are translated to English where a translation is available.
- Product column-name typos are standardized, including `product_name_lenght` and `product_description_lenght`.
- Geolocation records are deduplicated to the ZIP-code-prefix level using median latitude and longitude.
- Missing product measurements are filled with medians where appropriate.

4.2 Derived measures

The pipeline derives the following measures from order timestamps:

- `delivery_days`: purchase to customer delivery.
- `handling_days`: approval to carrier handoff.
- `transit_days`: carrier handoff to customer delivery.
- `delay_days`: actual delivery date minus estimated delivery date.
- `promise_buffer_days`: estimated delivery date minus purchase date.
- `is_late`: whether `delay_days` is greater than zero.
- `is_negative`: whether the review score is 1 or 2.

4.3 Review de-duplication

Some orders contain multiple review records. The preprocessing pipeline sorts reviews by score and retains one review per order, preserving the highest score as the documented conservative rule. The resulting review table is used for order-level satisfaction analysis.

4.4 Order-level aggregation

Order items are aggregated to order grain with item count, seller count, total price, total freight, freight ratio, primary seller, all seller IDs, category count, and primary category. Payment records are aggregated with payment method count, dominant payment type, maximum installments, and total payment value.

4.5 Data-quality treatment

Impossible timestamp relationships are flagged rather than silently discarded. The analysis also retains order statuses outside `delivered` and filters them explicitly for each question. Undelivered orders and missing delivery dates are treated as a separate analytical group where relevant.

5. Analytical Approach

The analysis combines descriptive statistics, grouped comparisons, concentration analysis, regional rankings, and robustness checks.

5.1 Population definitions

- Delivery-time analysis focuses primarily on delivered orders with valid delivery and promised-date values.
- Satisfaction analysis uses reviewed orders.
- Negative experience is defined as a review score of 1 or 2.
- Regional rankings use minimum order thresholds to avoid overinterpreting very small groups.
- Seller rankings use minimum order thresholds and distinguish raw complaint volume from complaint rate and excess complaints.

5.2 Main comparisons

The main explanatory comparison is between orders delivered on time and orders delivered after the estimated date. Additional comparisons examine:

- Early, on-time, and late delivery buckets.
- Absolute delivery duration among on-time orders.
- Lateness duration among late orders.
- Handling time versus carrier transit time.
- Customer states and seller-state to customer-state routes.
- Seller concentration, seller size, and seller performance.
- Product characteristics, freight ratio, and category revenue versus negative-review rate.

Where available, the analyses show group counts and confidence intervals. Small groups should be interpreted as investigation signals rather than final performance judgments.

6. Exploratory Analysis

The consolidated EDA report records 99,441 total orders, 98,673 reviewed orders, 96,478 delivered orders, and 95,832 delivered and reviewed orders. There are 96,096 unique customers and 3,088 unique sellers.

The review distribution is polarized: 57.84% of reviews are five stars and 11.47% are one star. Because one-star reviews alone are more common than two- and three-star reviews combined, the report uses the negative-review rate as the primary dissatisfaction measure rather than relying only on the mean score.

The overall negative-review baseline is 14.63% across reviewed orders. This baseline is useful for comparing regions, sellers, routes, and categories, but it should always be reported with the relevant population and sample size.

7. Key Findings

7.1 Missing the promised date is the clearest customer-experience breakpoint

The delivery analysis identifies a pronounced promise cliff:

- On-time orders have an approximate negative-review rate of 9.2%.
- Late orders have an approximate negative-review rate of 62.3%.
- Late orders are approximately 6.7% of deliveries but account for approximately 32.5% of negative reviews.
- The negative-review rate increases sharply after the promised date is missed, especially once lateness exceeds three days.

This result suggests that promise accuracy and exception management should be prioritized before broad attempts to reduce every delivery by the same number of days.

7.2 Slow delivery matters, but less abruptly than a broken promise

Among orders that remain on time, negative-review rates rise gradually as absolute delivery duration increases. Among late orders, negative-review rates rise much more sharply with additional lateness. The operational implication is that a customer may tolerate a longer wait when the promise is met, but reacts strongly when the marketplace misses the commitment.

Handling time and transit time are also different levers. Seller handling time is more directly controllable by the seller, while transit time is more dependent on the logistics network and route.

7.3 Multi-seller orders require better measurement

An order-level delivery status can hide the fact that an order contains products from multiple sellers. Customers may experience separate shipments even when the order is represented by one delivery event. The multi-seller analysis should therefore be presented as a measurement and tracking hypothesis, not as proven causality.

The recommended product change is to track seller legs or parcels separately, expose the status of each component to the customer, and evaluate satisfaction at both order and shipment level.

7.4 Seller concentration creates a targeted intervention opportunity

The seller analysis finds that the top 5% of sellers account for approximately 53.0% of orders and 45.8% of GMV, but approximately 54.6% of negative reviews. The worst 10% of sellers meeting the minimum order threshold account for approximately 14.1% of bad orders and 6.9% of GMV, or roughly twice as many bad orders per GMV point.

Seller size alone is not a reliable risk marker: the difference in average negative-review rate across seller-size groups is small. Seller interventions should therefore use context-adjusted performance, persistence over time, and minimum sample sizes rather than raw order volume.

7.5 Regional and route variation identifies where to investigate

Negative-review rates vary across customer states and seller-to-customer routes. The highest-risk rankings include both large routes and smaller routes, which is why order thresholds are required. Route analysis can distinguish a broad seller problem from a network or geography problem and can inform carrier capacity, promise-setting, and exception-management decisions.

Regional rankings should be treated as prioritization signals. They do not prove that geography itself causes dissatisfaction because state and route performance may also reflect product mix, distance, seller mix, and promised-date quality.

7.6 Product and freight characteristics are secondary levers

Product and pricing analysis examines price, freight ratio, weight, volume, photo count, description length, item count, and category revenue. The analysis suggests that listing quality and freight provide additional levers, but delivery remains the strongest observed satisfaction factor.

High-revenue categories with above-baseline negative-review rates are especially useful targets because improvements can protect both customer experience and commercial value. These relationships are associative and should be validated with controlled operational tests.

8. Recommended Actions

Priority 1: Protect the delivery promise

- Recalibrate estimated delivery dates using route, seller, season, and carrier performance.
- Monitor promise-miss rate as a primary operational KPI.
- Create early-warning alerts for orders likely to miss the promise.
- Contact customers proactively when a miss becomes likely and provide a revised estimate.
- Track the percentage of late orders that receive an exception intervention.

Priority 2: Separate seller, carrier, and route accountability

- Decompose total delivery time into handling and transit time.
- Give sellers feedback on controllable handling delays.
- Review carriers and routes with persistent transit delays.
- Compare seller performance with expected performance for its category and delivery context.
- Use minimum volume thresholds and persistence checks before imposing penalties.

Priority 3: Improve multi-seller visibility

- Store shipment-level or seller-leg delivery events.
- Show customers the status and expected date for each parcel.
- Report both order-level and parcel-level on-time rates.

- Test whether clearer tracking reduces negative reviews for multi-seller orders.

Priority 4: Target high-value dissatisfaction

- Prioritize high-GMV categories with above-baseline negative-review rates.
- Audit listings with weak photo coverage or incomplete descriptions.
- Review extreme freight ratios for pricing transparency and customer expectations.
- Use seller coaching before delisting, except where severe or persistent performance risk warrants escalation.

Priority 5: Establish an intervention measurement loop

For each intervention, define a baseline, treatment population, comparison group, and review period. Track negative-review rate, late rate, promise-miss rate, customer contacts, refunds, and GMV. A future experiment should test whether more accurate promises and proactive exception messaging reduce negative reviews without reducing conversion or seller participation.

9. Limitations and Risks

- The dataset is observational; associations should not be interpreted as causal effects.
- Delivery analysis is affected by missing dates and the exclusion of undelivered orders from some timing comparisons.
- Review de-duplication requires a documented rule and can change reported rates.
- A single order-level delivery event may not represent the experience of every parcel in a multi-seller order.
- State and route rankings can be unstable for low-volume groups.
- Some late orders in the 30+ day bucket may reflect unresolved data-quality issues and should not be interpreted as a clean operational segment.
- The seller analysis is strongest when minimum order thresholds, context adjustment, and persistence checks are applied together.
- The available Sahib seller-behaviour artifact is a PDF without readable source outputs in the repository; its methods and findings are not treated as independently verified in this report.

10. Reproducibility

From the repository root:

```shell
python3 -m pip install -r requirements.txt
python preprocess.py
streamlit run app.py
```

Run preprocessing before analysis so that the files in `processed_data` are regenerated from the raw dataset. The Streamlit portal provides access to the raw-data overview, cleaned-data overview, EDA, business questions, analyses, and deliverables. Individual notebooks can be opened and rerun from the `analysis/` directory.

Appendix A: Project Resources

A.1 Project Resources

Presentation: https://github.com/DSAshv/DVDProject/blob/master/PRESENTATION.pdf
Project management page: https://dvd-project.streamlit.app/
Interactive dashboard: https://dvd-project.streamlit.app/?page=dashboard
Repo: https://github.com/DSAshv/DVDProject

A.2 Team analyses and summaries

Source	**Contents**
`analysis/Yash/Delivery_Performance_Analysis.ipynb`	Promise-cliff, delivery speed, undelivered-order, review-timing, and robustness analysis.
`analysis/Anushka/Regional_Logistics_Analysis.ipynb`	Customer-state and seller-state to customer-state route analysis.
`analysis/Kannan/Seller_Performance_Analysis.ipynb`	Seller concentration, context-adjusted seller performance, persistence, and intervention analysis.
`analysis/Ashwanth/Product_Pricing_Analysis.ipynb`	Product, pricing, freight, listing-quality, and category-revenue analysis.
`analysis/Ashwanth/Product_Pricing_Analysis.html`	Rendered HTML version of the product-pricing analysis.
`analysis/Sahib/Sahib_Seller_Behaviour_Investment_Report.pdf`	Seller-behaviour and investment report; source methods are not currently machine-readable in the repository.
`analysis/individual_summaries/Yash.md`	Summary of delivery-performance methods and conclusions.
`analysis/individual_summaries/Anushka.md`	Summary of regional-logistics methods and conclusions.
`analysis/individual_summaries/Kannan.md`	Summary of seller-performance methods and conclusions.

Source	Contents
`analysis/individual_summaries/Ashwanth.md`	Summary of product-pricing methods and conclusions.
`analysis/individual_summaries/Sahib.md`	Evidence and verification status of the seller-behaviour report.

Appendix B: Definitions

Term	Definition
Negative review	Review score of 1 or 2.
On-time order	Delivered on or before the normalized estimated delivery date.
Late order	Delivered after the normalized estimated delivery date.
Delay days	Actual delivery date minus estimated delivery date; positive values indicate lateness.
Handling time	Time from order approval to carrier handoff.
Transit time	Time from carrier handoff to customer delivery.
GMV	Total item price used as the order or seller merchandise-value measure in the analysis.
Multi-seller order	An order containing items from more than one seller.